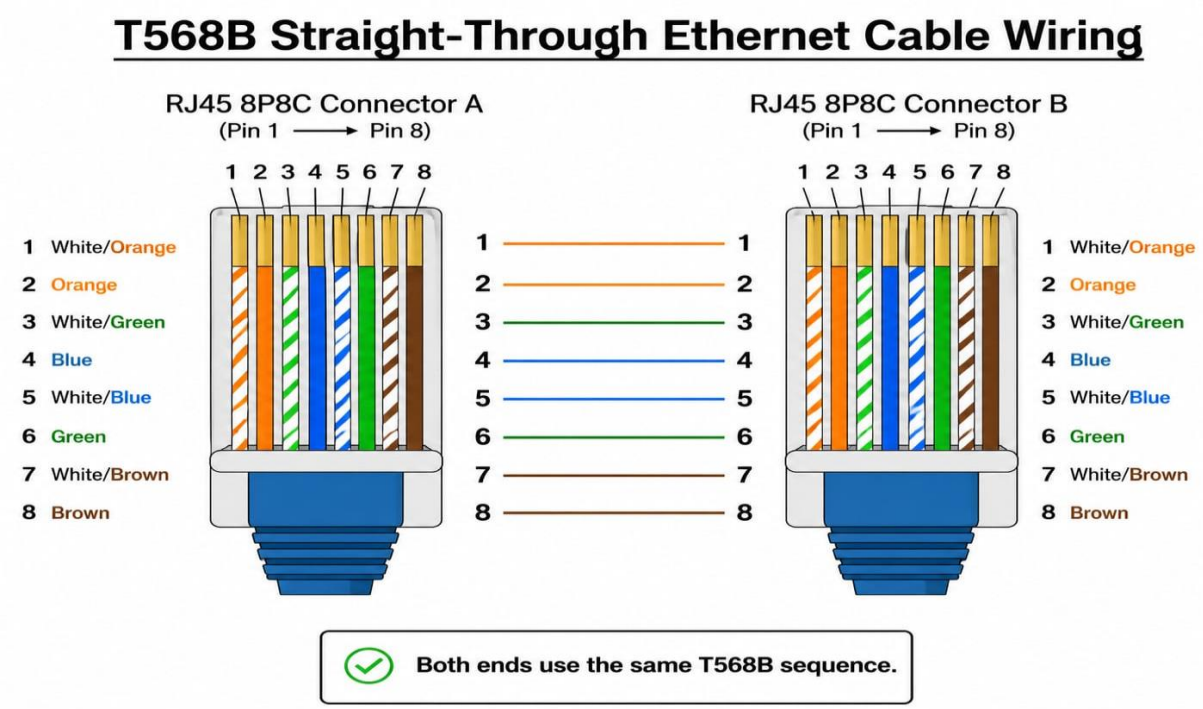
Q1. Write 3 differences between Ethernet cables and Fiber Optic cables.

Point	Ethernet Cable	Fiber Optic Cable
1. Data transmission	Uses electrical signals.	Uses light signals.
2. Speed and distance	Suitable for LAN connections over Shorter distances.	Suitable for very high-speed and long-distance connections.
3. Cost	Generally cheaper and easier to install.	Generally more expensive and requires specialized tools.
Example	Connecting a computer to a home router.	Connecting buildings in a college campus network.

Q2. Draw and Label a T568B Straight-Through Ethernet Cable



T568B wire colour sequence:

Pin Number	Wire Colour
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

Explanation: A straight-through Ethernet cable uses the same wiring standard at both ends. In this assignment, both ends use T568B.

Q3. Steps to Attach an RJ45 Connector to a CAT6 Cable

1. Gather a CAT6 cable, RJ45 connectors, an RJ45 crimping tool, and a cable stripper.

2. Cut the cable to the required length.

3. Carefully remove a small portion of the outer jacket.

4. Separate and arrange the eight wires in T568B colour order.

5. Straighten the wires and trim them evenly.

6. Insert the wires fully into the RJ45 connector, ensuring each wire reaches its correct position.

7. Use the crimping tool to secure the connector.

8. Test the cable using a LAN cable tester.

Result: A straight-through CAT6 Ethernet cable is ready for network use after successful testing.

Q4. Gaming Café with 10 PCs

I would use CAT6 Ethernet cables to connect all 10 gaming PCs to a Gigabit network switch. Ethernet cables are affordable, easy to install, and provide stable, low-latency connections for LAN gaming. I would use an RJ45 crimping tool to attach the connectors to the cables. Fiber optic cables are better suited for long-distance or high-speed backbone links, but they are generally unnecessary for connecting PCs within a small gaming café.